

# 20210331-131117-0500–My talk: Schur-Weyl Duality

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## Contents

*[2021-03-31 Wed 13:11]*

**tag** Schur-Weyl duality, double commutant theorem, representation theory

Disclaimer: this talk does not talk about Schur-Weyl duality for the quantum affine nor the quantum toroidal setting.

## Reference

- [A] Drinfeld Functor and Finite-Dimensional Representations of Yangian-[Tomoyuki Arakawa]
- [C and S] Quantum affine algebras and affine Hecke algebras-[Vyjayanthi Chari and Andrew Pressley]-[arXiv:q-alg-9501003]
- [MNO] Yangians and classical Lie algebras-[Molev and Nazarov and Olshanski]
- [S] Schur Weyl Duality for Quantum Groups-[Yi Sun]
- [GW] Symmetry, Representations, and Invariants-[R. Goodman and N. Wallach]

## Talk

### Classical

### Motivation

### ATTACH

- Decomposition of finite group representations

Motivation - Decomposition of <sup>group-</sup>representations

eg.  $V \otimes V \cong \underbrace{\text{Sym}_2 V}_{\cong \mathbb{S}_{\square} V} \oplus \underbrace{\Lambda^2 V}_{\cong \mathbb{S}_{\blacksquare} V}$

$V \otimes V \otimes V \cong (\mathbb{S}_{\square\square} V)^{\oplus \dim \square} \oplus (\mathbb{S}_{\square\blacksquare} V)^{\oplus \dim \blacksquare} \oplus (\mathbb{S}_{\blacksquare\blacksquare} V)^{\oplus \dim \blacksquare}$

$V^{\otimes k} \in \bigoplus_{\lambda \vdash k} (\mathbb{S}_{\lambda} V)^{\oplus \dim(\lambda)}$

Check out the character table.  
Sage: `SymmetricGroup(n).character_table();`  
or Hook length formula

- For  $\dim(\lambda)$ , check out the character tables for the symmetric groups, or use the hook length formula.

```
for n in [1 .. 4]:  
    print(n);  
    SymmetricGroup(n).character_table();  
    print("");
```

```
1  
[1]  
  
2  
[ 1 -1]  
[ 1  1]
```

3

$$\begin{bmatrix} 1 & -1 & 1 \\ 2 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix}$$

4

$$\begin{bmatrix} 1 & -1 & 1 & 1 & -1 \\ 3 & -1 & -1 & 0 & 1 \\ 2 & 0 & 2 & -1 & 0 \\ 3 & 1 & -1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

### Schur-Weyl duality (classical)

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- The above statement can be refined to be a duality statement between the symmetric groups  $S_k$  and the reductive algebraic groups  $GL(n; \mathbb{C})$  or  $SL(n; \mathbb{C})$ .

This is the classical Schur-Weyl duality

Schur-Weyl: Let  $V$  be a finite dim<sup>l</sup> <sup>(or Hecke length formula)</sup> vector space,  $n \in \mathbb{N}$ . Then as a  $S_n \otimes GL(n, \mathbb{C})$  module,

$$\otimes^k(\mathbb{C}^n) \simeq \bigoplus_{\substack{\lambda \vdash n \\ \ell(\lambda) \leq n}} (S_\lambda V) \boxtimes L_\lambda$$

where  $L_\lambda$  is the highest wt module of  $SL(n, \mathbb{C})$   
 i.e/ weight =  $(\lambda_1 - \lambda_2)\omega_1 + (\lambda_2 - \lambda_3)\omega_2 + \dots + (\lambda_{n-1} - \lambda_n)\omega_{n-1}$   
 and  $zI \in GL(n, \mathbb{C})$  acts as  $z^k$ .

(ref. Goodman & Wallach thm 5.5.22)  
 thm 9.1.2

- Note that when  $n \geq k$ , all irreps of  $S_k$  show up in the direct sum!

## Sketch of proof

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- As the proofs of the quantum cases all seem to be based on this fact, I figure it would be nice to know a sketch of its proof.
- The idea is to use a general duality theorem, that requires the two algebras to be double commutant to each other.

Double Commutant + Duality

- First, we have the classical Schur's double commutant theorem:

Schur's Theorem (Double Commutant) [Goodman + Wallach Thm 4.2.10]

By the two actions, we have  $S_K \hookrightarrow \text{End}(\otimes^n \mathbb{C}^n) \hookrightarrow \text{GL}(n, \mathbb{C})$

For any subalg  $A \subseteq \text{End}(\otimes^n \mathbb{C}^n)$ , denote its commutant be

$$A' := \{ X \in \text{End}(\cdot) \mid ax = xa \quad \forall a \in A \}$$

Then  $(S_K)' = \text{GL}(n, \mathbb{C})$  and  $S_K = (\text{GL}(n, \mathbb{C}))'$ .

- This allows us to use the following general duality theorem.

Thm (General Duality) [G+W, Thm 4.2.1]

Let  $G \subseteq \text{GL}(n, \mathbb{C})$  be a reductive linear algebraic group.

Let  $(\rho, L)$  be a locally regular rep of  $G$  w/ countable dim.

Let  $R$  be a subalg of  $\text{End}(L)$  s.t.  $R$  acts irreducibly on  $L$ . Suppose  $\rho(g) \cdot r \rho(g)^{-1} \in R \quad \forall g \in G, r \in R$ , so  $G \curvearrowright R$ , and suppose that this action is also (or) regular.

Then

- 1)  $L \simeq \bigoplus_{\lambda \in \text{Spec}(R)} E^\lambda \otimes F^\lambda$  as an  $R \otimes G$  module  
 $E^\lambda \in \text{Hom}_R(F^\lambda, L)$
- 2)  $E^\lambda$  is irrep of  $R$ , and  $E^\lambda \leftrightarrow F^\lambda$  is a bijection on irreps of both that occur in  $L$ .

- Indeed,

By these theorems, w/

p3/4

$$L = \otimes^k(\mathbb{C}^n)$$

$$G = GL(n, \mathbb{C})$$

$$R = \text{End}(L).$$

We have  $R^G = [S_k]$ , so.

$$\otimes^k(\mathbb{C}^n) \simeq \bigoplus_{\lambda \in (\otimes^k \mathbb{C}^n \cap GL(n, \mathbb{C}))} E^\lambda \boxtimes F^\lambda$$

as an  $S_k \otimes GL(n, \mathbb{C})$ -module.

Further analysis on weights and characters tells

us that 
$$\otimes^k(\mathbb{C}^n) \simeq \bigoplus_{\substack{\lambda \vdash k \\ s(\lambda) \leq n}} (S_\lambda V) \boxtimes L_\lambda.$$
 #

#### • Aside

1. If you see an essentially different proof, please let me know.
2. Before preparing this talk, I hoped the quantum cases can be proved by some very (more) general duality theorem. But all accounts I have do not mention such method.

The general duality theorem I have here does require one side to be a reductive group, and the vector space to be a locally regular representation.

#### Remarks

- We have a coincidence -  $S_k$  is the Weyl group of type A.

This is merely a coincidence, as in type B, C, D, we don't have W on the other side. Instead, one should look for Brauer algebras and partition algebras [Halverson - 1605.06543].

- To pass to the quantum case, we first replace  $G = \mathrm{SL}(n; \mathbb{C})$  by  $\mathfrak{sl}_n$  and then by  $U(\mathfrak{sl}_n)$  – This does not change the representation theory.

Then we deform and get the quantum group  $U_q(\mathfrak{sl}_n)$ , for  $q$  not a root of unity.

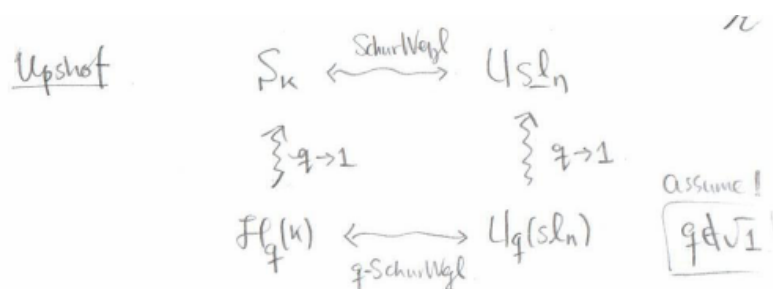
## Quantum (finite type)

In this section, we follow Yi Sun's note on Schur duality.

### $U_q(\mathfrak{sl}_n)$ and Hecke algebra

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- Upshot



Schur-Weyl duality extends to  $U_q(\mathfrak{sl}_n)$ !

- Compare the  $\mathrm{Rep}(U(\mathfrak{sl}_n))$  and  $\mathrm{Rep}(U_q(\mathfrak{sl}_n))$ 
  1. They are both monoidal, with isomorphic Grothendieck rings.
  2. However, they differ as monoidal categories (c.f. 6j symbols or F-symbols).
  3. As braided categories, they differ even more.

Recall that  $\begin{pmatrix} V \otimes W \rightarrow W \otimes V \\ v \otimes w \mapsto w \otimes v \end{pmatrix}$  is a morphism in  $\text{Rep}(U(\mathfrak{sl}_n))$ , but NOT IN  $\text{Rep}(U_q \mathfrak{sl}_n)$ .

This boils down to the fact that  $U(\mathfrak{sl}_n)$  is cocommutative, whereas  $U_q \mathfrak{sl}_n$  isn't by design. Nevertheless,  $V \otimes W \cong W \otimes V$  in both case, and in the  $q \neq 1$  case it's realized by nontrivial  $R$ -matrices:  $V \otimes W \xrightarrow[\sim]{\hat{R}_q} W \otimes V$ .

such that  $\hat{R}^2 \neq 1$ .  $\frac{p^2}{2}$

$$\hat{R}^{23} \hat{R}^{12} \hat{R}^{23} = \hat{R}^{12} \hat{R}^{23} \hat{R}^{12} \quad (\text{YB})$$

$$\hat{R} |_{\mathbb{C}^n \otimes \mathbb{C}^n}$$

$$\begin{aligned} & q \sum_i E_{ii} \otimes E_{ii} \\ & + \sum_{i \neq j} E_{ii} \otimes E_{jj} \\ & + (q - q^{-1}) \sum_{i > j} E_{ij} \otimes E_{ji} \end{aligned}$$

$$\begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} = \begin{array}{c} \diagdown \diagup \\ \diagup \diagdown \end{array}$$

4. The fact that  $R^2 \neq 1$  stops  $S_k$  from acting on  $\otimes^k(\mathbb{C}^n)$ .. instead, we have a Hecke algebra action:

$$H_q(k) := \left\langle T_1, T_2, \dots, T_{k-1} \mid \begin{array}{l} (T_i - q^{-1})(T_i + q) = 0 \\ T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1} \\ [T_i, T_j] = 0 \text{ if } |i-j| \neq 1 \end{array} \right\rangle$$

by  $T_i \mapsto \hat{R}^{i, i+1} \mid$   
 $(\otimes^k \mathbb{C}^n)$

Double commutant for  $U_q(\mathfrak{sl}_n)$  and  $H_q(m)$

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(Double commutant)  
Theorem 2.6(a) If  $q \notin \sqrt{1}$ , and if  $n > m$ , we have  
 that the image of  $U_q(\mathfrak{sl}_n)$  and the image of  $H_q(m)$   
 in  $\text{End}_{\mathbb{C}}(V^{\otimes m})$  are commutants of each other,  
 where  $V$  denotes the std rep of  $U_q(\mathfrak{sl}_n)$ .

Proof

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- notation

$$\begin{aligned} \text{Denote } A_q &= \text{image}(U_q(\mathfrak{sl}_n)) \subseteq \text{End}(V^{\otimes m}) \\ B_q &= \text{image}(H_q(m)) \subseteq \text{End}(V^{\otimes m}). \end{aligned}$$

We will show that  $B_q = A_q'$   
 and that  $A_q = B_q''$ .

- 1.  $B_q = A_q'$

1.1)  $B_q \subseteq A_q'$ : Immediate by def of the action.. one is from left & one is from right.

1.2)  $B_q = A_q'$ : We will count dimensions. From the classical case,

we have  $B_{q=1} = (A_{q=1})'$ .

It remains to show that  $\dim A_q = \dim A_1$   
and  $\dim B_q = \dim B_1$

$$- \dim(A_q) = \dim(A_1)$$

•  $q \neq \sqrt{1} \Rightarrow U_q(\mathfrak{sl}_n)$  semiSimple ref 12  
Kassel - On the  
Group VII.2.2.  
(for  $\mathfrak{sl}_2$ )  
see also Kassel  
QG. §5.17  
so  $V^{\text{mod}} \simeq \bigoplus_a M_a$  "for finite dim mod modules"  
 $U_q \text{ mod } a$  (inep)

and  $A_q \simeq \bigoplus_a \mathbb{C}_a$  by schur's lemma.

• But the Grothendieck groups of  
 $\text{Rep}(U_q \mathfrak{sl}_n)$  and  $\text{Rep}(U_1 \mathfrak{sl}_n)$   
are the same (ref?),

so  $\dim(A_q) = \dim(A_1)$ .

or  
Curtis - On  
Lusztig's isom  
thm for  
Hecke algs

henry:  $U_q(\mathfrak{sl}_2)$  is not semisimple take evaluation representations...  
etc

–  $\dim(B_q) = \dim(B_1)$

2)  $\dim B_q = \dim B_1$

•  $q \notin \sqrt{1} \Rightarrow H_q(m) \simeq_{\text{alg}} \mathbb{C}[S_m]$  (ref. <sup>Jimbo - A q-analogue of  $U(\mathfrak{gl}(N))$ , proof of proposition 3</sup>).

•  $H_q(m)$  acts on  $V^{\otimes m}$  faithfully (why?)

and  $\mathbb{C}[S_m]$  too (trivial), so

$\dim B_q = \dim B_1$ .  
completes 1).

• 2.  $A_q = A_q''$

1)  $A_q = A_q''$

Since  $U_q(\mathfrak{sl}_n)$  is semisimple,  $V^{\otimes m}$  is a completely reducible  $U_q(\mathfrak{sl}_n)$  module.

By a general algebra theorem (ref. <sup>Goodman & Wallach Thm 4.1.13</sup>),

$A_q = A_q''$ .

# This completes the proof of thm 2.6 (a).

Duality for  $U_q(\mathfrak{sl}_n)$  and  $H_q(m)$

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- q-Schur-Weyl Duality

Theorem 2.6(b) If  $q \notin \sqrt{1}$ , we have that

$$V^{\otimes m} = \bigoplus_{\substack{\lambda \vdash m \\ l(\lambda) \leq n}} S_{\lambda}^q \otimes L_{\lambda}$$

as a  $\mathcal{H}_q(m) \otimes U_q(\mathfrak{sl}_n)$  module.

**Proof**

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As mentioned in the previous proof, finite dim modules of  $U_q(\mathfrak{sl}_n)$  are semisimple. So,

$$V^{\otimes m} = \bigoplus_{\lambda \in \hat{U}_q} \underbrace{\text{Hom}_{U_q}(L_{\lambda}, V^{\otimes m})}_{\leftarrow \text{vect}} \otimes \underbrace{L_{\lambda}}_{U_q \text{ mod}}$$

Obviously, this decomposition is as a  $\mathcal{H}_q(m) \otimes U_q(\mathfrak{sl}_n)$ -module as well.

It remains to prove that  $\text{Hom}_{U_q}(L_{\lambda}, V^{\otimes m})$  as an  $\mathcal{H}_q(m)$ -module is the  $q$ -deformation of the  $[S_m]$  module  $S_{\lambda}$ .

This is claimed to be true in [Visan].. but <sup>IMHO</sup> a complete proof should ~~have~~ be done by writing down an explicit map mimicking the classical case. (omit)<sub>†</sub>

## Corollary

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Corollary 2.7. For  $n > m$ , the functor  $\text{Rep}(\mathfrak{sl}_q(m)) \xrightarrow{\text{FS}_q} \text{Rep}(U_q(\mathfrak{sl}_n))$   
 $W \mapsto \text{Hom}_{\mathfrak{sl}_q(m)}(W, V^{\otimes m})$   
 is an equivalence of cats  
 between  $\text{Rep}(\mathfrak{sl}_q(m))$  and the full subcat  
 of weight  $m$  reps of  $U_q(\mathfrak{sl}_n)$ .  
 $\nwarrow$  meaning those showing up in  $V^{\otimes m}$ .

### • Proof

By ss., we have  $V^{\otimes m} = \bigoplus_{\substack{\lambda \vdash m \\ \ell(\lambda) \leq n}} \mathbb{S}_\lambda^q \otimes \text{Hom}_{\mathfrak{sl}_q(m)}(\mathbb{S}_\lambda^q, V^{\otimes m})$   
 as an  $\mathfrak{sl}_q(m)$ -module.  
 By 2.6(b), we have  $\text{Hom}_{\mathfrak{sl}_q(m)}(\mathbb{S}_\lambda^q, V^{\otimes m}) \cong_{(U_q^{\text{mod}})} L_\lambda$ ,  
 and also that  $\text{FS}_q$  is ess. onto the "wt  $m$ " subcat.  
 Since both  $\mathbb{S}_\lambda^q$  and  $L_\lambda$  are simple,  $\text{FS}_q$  is indeed  
 fully faithful. #

## Quantum (Yangian)

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### • reference

ref: [Yi Sun — Schur-Weyl duality for Quantum groups §3]  
 [Molev, Nazarov, Olshanski — Yangians and Classical  
 Lie algs §1 + §2]  
 [Arioka — Drinfeld functor & fin dim'd reps of Yang-  
 Yang-Baxter equation]

Informal talk note. compact.

## Yangian of type A

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- Yang Baxter equation (with a spectral parameter)

$$R_{12}(u_1 - u_2) R_{13}(u_1 - u_3) R_{23}(u_2 - u_3) \\ \parallel \\ R_{23}(u_2 - u_3) R_{13}(u_1 - u_3) R_{12}(u_1 - u_2).$$

for  $R(u) \in \text{End}(\mathbb{C}^n \otimes \mathbb{C}^n) \otimes \mathbb{C}(u).$

This is solved in terms of the Yangians of  $\mathfrak{gl}_n = Y(\mathfrak{gl}_n).$

- $Y(\mathfrak{gl}_n)$

Def  $(Y(\mathfrak{gl}_n))$  As an algebra, it is

$$\left[ \begin{array}{l} \text{MNO, 1.1} \\ 1.8 \end{array} \right] \left[ \begin{array}{l} \text{S. 3.1} \end{array} \right] \left\langle t_{ij}^{(k)} \mid \begin{array}{l} k=0, 1, 2, \dots \\ 1 \leq i, j \leq n \end{array} \right\rangle$$

where  $R = 1 - \frac{P}{u}$

$$t(u) = \sum_{ij} t_{ij}(u) \otimes E_{ij}$$

$$t_{ij}(u) = \delta_{ij} u^{-1} + \sum_{k \geq 1} t_{ij}^{(k)} u^{-k-1}$$

Thm Let  $i, j, k$  pairwise distinct.

- Theorem  $Y(\mathfrak{gl}_n)$  solves the YB equation with one spectral parameter.
- Some remarks
  1.  $Y(\mathfrak{gl}_n)$  has a Hopf algebra structure

$$\Delta(t_{ij}(u)) = \sum_{a=1}^n t_{ia}(u) \otimes t_{aj}(u)$$

$$S(t(u)) = t(u)^{-1} \quad \left[ \begin{array}{l} \text{MNO} \\ 1.28 \end{array} \right]$$

2.  $U(\mathfrak{gl}_n)$  embeds into  $Y$  as a Hopf algebra

$$\begin{array}{ccc} U(\mathfrak{gl}_n) & \xrightarrow{\text{Hopf}} & Y(\mathfrak{gl}_n) \\ E_{ij} & \mapsto & t_{ij}^{(1)} \end{array} \quad [MNO \text{ 1.17}]$$

3. For type A, there's an exceptional algebra morphism going the other side.

$$\begin{array}{ccc} U(\mathfrak{gl}_n) & \xleftarrow[\text{alg.}]{\text{ev}_a} & Y(\mathfrak{gl}_n) \\ \delta_{ij} + \frac{E_{ij}}{a-a} & \longleftarrow & t_{ij}(u) \end{array} \quad [MNO \text{ 1.16}]$$

( $a \in \mathbb{C}$ )

This map seems to be exceptional for type A [MNO §0].

4.  $Y(\mathfrak{gl}_n)$  is a deformation for the current lie algebra  $U(\mathfrak{gl}_n[x])$ .

- $Y(\mathfrak{sl}_n)$

Let  $f = 1 + f_1 u^{-1} + f_2 u^{-2} + \dots \in \mathbb{C}[[u^{-1}]]$

Then  $\mu_f: Y(\mathfrak{gl}_n) \longrightarrow Y(\mathfrak{gl}_n)$

$$t(u) \mapsto f(u)t(u)$$

is an automorphism.

Def  $Y(\mathfrak{sl}_n) := \{x \in Y(\mathfrak{gl}_n) \mid \mu_f(x) = x \ \forall f \in \mathbb{C}[[u^{-1}]] \text{ w/ } f_0 = 1\}$

[§.3.2] This is a Hopf subalgebra of  $Y(\mathfrak{gl}_n)$ .  
[MNO §2]

- Remarks

1. We have compatible embeddings

$$\begin{array}{ccc}
 U(\mathfrak{sl}_n) & \hookrightarrow & Y(\mathfrak{sl}_n) \\
 \downarrow & \supseteq & \downarrow \\
 U(\mathfrak{gl}_n) & \hookrightarrow & Y(\mathfrak{gl}_n)
 \end{array}$$

2. Representations of weight  $\mathbf{m}$

Say a rep of  $Y(\mathfrak{sl}_n)$  is of weight  $\mathbf{m}$   
 if it is of wt  $\mathbf{m}$  as a  $U(\mathfrak{sl}_n)$ -rep.  
 $\uparrow$   
 recall this means that it occurs  
 in  $(\mathbb{C}^n)^{\otimes \mathbf{m}}$ .

3. Structure of  $Y(\mathfrak{gl}_n)$

$$Y(\mathfrak{sl}_n) \simeq \frac{Y(\mathfrak{gl}_n)}{(\det(t_{ii}) - 1)}$$

$$Y(\mathfrak{gl}_n) \simeq \text{Center} \otimes Y(\mathfrak{sl}_n)$$

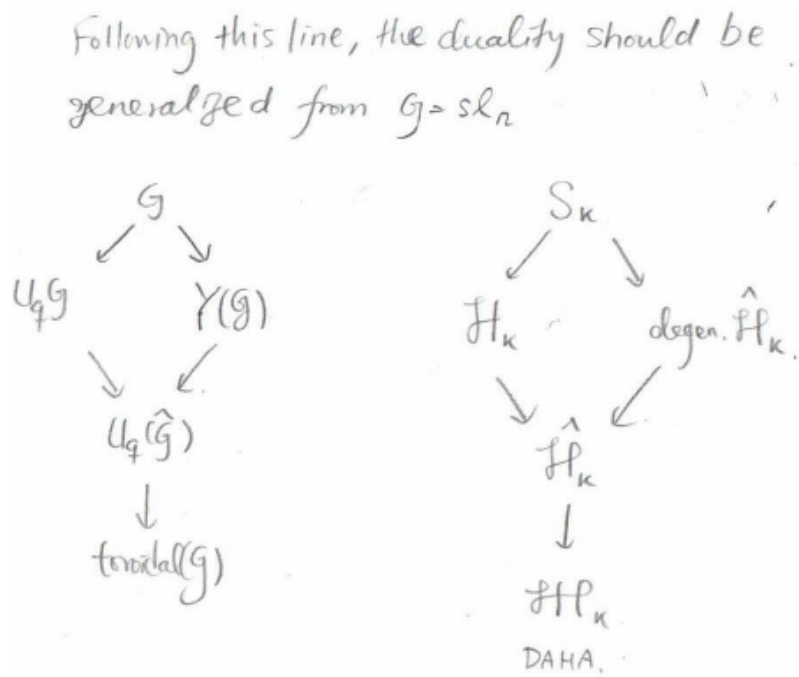
**Schur-Weyl duality for  $Y(\mathfrak{sl}_n)$**

NONE. See the final section.

## Beyond (conclusion)

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- The stories that follow



- Reference

- For  $Y(\mathfrak{g})$ , I haven't seen a complete account.
- For  $U_q(\hat{\mathfrak{g}})$ , [Chari + Pressley - Quantum affine algs and affine Hecke algs] could be a complete account. According to [A], its technique can be used to produce a proof for the Yangian case.
- For  $\text{toroidal}(\mathfrak{g})$ , the standard account is [Varagnolo, Vasserot - Schur duality in the toroidal setting] ← I haven't checked its details

- In the rest of this talk, we will give the Drinfeld functor that embeds  $\text{Rep}(\text{degenerated AHA})$  into  $\text{Rep}(Y(\mathfrak{gl}_n))$ .

### 1. (Degenerated AHA)

$\frac{1}{3}$

(Degen. affine Hecke algebra)  
type  $A_{k-1}$ .

$$\Lambda_k \underset{(\text{vect})}{\simeq} \mathbb{C}[S_k] \otimes \text{Sym}(\eta_{(S_k)}) \xrightarrow{\quad} \text{Poly}(\eta_{S_k}^*)$$

$$\underset{\text{alg}}{\simeq} \left\langle \begin{array}{c} s_1, \dots, s_{k-1} \\ x_1, \dots, x_k \end{array} \right\rangle$$

$$\begin{array}{l} s_i^2 = 1 \\ s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1} \\ [x_i, x_j] = 0 \\ s_i x_i - x_{i+1} s_i = 1 \\ [s_i, s_j] = [s_i, x_j] = 0 \text{ if } |i-j| \neq 1 \end{array}$$

For a review of its representation, see [A. section 1.4.]

### 2. (Drinfeld functor)

Drinfeld Functor [A. §2]

For a left  $\Lambda_k$  module  $M$ , we will give a  $Y(\mathfrak{gl}_n)$ -module.

Define  $D_k(M) := M \otimes_{\text{vect.}} (\mathbb{C}^n)^{\otimes l} / \sum_{i=1}^{k-1} I_m(s_i + 1)$

where  $I_m(s_i + 1) = (s_i + 1)(M \otimes (\mathbb{C}^n)^{\otimes l})$   
 $s_i$  acts on  $M$  by  $S_k \mapsto A_k$ , and on  $(\mathbb{C}^n)^{\otimes l}$  by permutation.

### 3. $Y(\mathfrak{gl}_n)$ action

$Y(\mathfrak{gl}_n)$  acts on  $D_k(M)$  by [A. (2.2.1)]

$$t_{ab}^{(d)} \mapsto \sum_{i=1}^n (y_i)^d \otimes T_i(E_{ab}).$$

$$(y_i) := (x_i - \sum_{j < i} S_{ij})$$

"only acts on the  $i$ th of  $\mathbb{C}^n$  in  $(\mathbb{C}^n)^{\otimes K}$ "

#### 4. "State of the proof/theorem"

Theorem [A P.8]

$$\text{For } n > k, \text{ Rep}(A_k) \hookrightarrow \text{Rep}(Y(\mathfrak{sl}_n))$$

Proof See "[D2]", which is in Russian and of 2-page long. Actually, there's no proof in that paper, but one can deduce a proof from "[CP2]", which works for an even more general case.

$$\text{Rep}(AHA_k) \hookrightarrow \text{Rep}(U_q(\widehat{\mathfrak{sl}_n})).$$